

STAT 412 HW07 — Question 1

Problem

Suppose that an allergist wishes to test the hypothesis that at least 31% of the public is allergic to some cheese products. Explain how the allergist could commit a type I error and a type II error.

Solution

The platform wording treats the tested claim as “31% or more.” A type I error means concluding the claim is true when it is actually false:

conclude $p \geq 0.31$ when actually $p < 0.31$.

A type II error means missing the claim when it is actually true:

conclude $p < 0.31$ when actually $p \geq 0.31$.

Final Answer

Type I: conclude 31% or more are allergic when actually less than 31% are allergic. Type II: conclude less than 31% are allergic when actually 31% or more are allergic.